

M4 — Internal Supply Sources (12%)

PAB = Prior PAB + Sched Receipts + Planned Receipts – Gross Reqs

Projected Available Balance carried period to period in the MRP record.

USE Every MRP record question. Row that drives net requirements.

Net Requirements = Gross Reqs – Sched Receipts – Prior PAB

Shortage that triggers a planned order release.

USE Only calculate when result > 0. Otherwise net req = 0.

ATP (Discrete) = MPS + Sched Receipts – Σ Cust Orders (to next MPS)

Uncommitted inventory in a specific bucket.

USE Order promising when MPS receipts are period-specific.

ATP (Cumulative) = Prior ATP + Period ATP

Rolling running total of ATP across buckets.

USE When shortages in one period can be covered by future supply.

M6 — Inventory (14%)

EOQ = $\sqrt{(2DS / H)}$

D = annual demand, S = order cost, H = annual holding cost per unit.

USE Fixed order quantity when demand is stable and independent.

Safety Stock = $Z \times \sigma \times \sqrt{LT}$

Z = service factor, σ = demand std dev per period, LT = lead time in periods.

USE Buffer against demand variability during lead time.

Reorder Point = $(d \times LT) + SS$

d = average demand per period, LT = lead time, SS = safety stock.

USE Continuous review (perpetual) inventory systems.

Inventory Turnover = COGS / Average Inventory

How many times inventory sells through per year.

USE Efficiency benchmark. Higher = leaner (but too high risks stockouts).

Days of Supply = Average Inventory / Daily Usage

How many days of demand current inventory covers.

USE Inverse view of turns. Easier for operational conversations.

M7 — Detailed Scheduling (12%)

Critical Ratio = Time Remaining / Work Remaining

Time until due date divided by lead time still needed.

USE Dispatch priority. CR < 1 = behind; = 1 = on time; > 1 = ahead.

Utilization = Actual Hours Worked / Available Hours

How much of available capacity was actually used.

USE Capacity performance monitoring against available time.

Efficiency = Standard Hours Earned / Actual Hours Worked

How productively the hours were spent vs standard.

USE Labor productivity benchmark against engineered standards.

Rated Capacity = Available $\times$ Utilization $\times$ Efficiency

Sustainable output level of a work center.

USE RCCP and detailed capacity planning.

Takt Time = Available Production Time / Customer Demand

Pace of production required to meet demand.

USE Line balancing in lean/pull environments.

M3 — Demand (12%)

Tracking Signal = CFE / MAD

Cumulative forecast error divided by mean absolute deviation.

USE Detect forecast bias. |TS| > 4 = investigate; > 7 = model broken.

EXAM DAY REMINDERS

150 questions · 3.5 hours · Pass = 300 · 84 sec/question

Z-FACTORS (memorize):

90% → Z=1.28 95% → Z=1.65 97% → Z=1.88

98% → Z=2.05 99% → Z=2.33 99.9% → Z=3.09

IF STUCK:

- Flag & skip — return with time budget
- Never leave blank (no guessing penalty)
- Watch units: annual vs period vs daily
- Read question TWICE before options